On the Identity Between the Sum of Cubes and the Square of the Sum

Dakshesh Nankani

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Proposition

For every positive integer n,

$$\sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k \right)^2.$$

Equivalently,

$$\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}.$$

We establish this identity through two independent approaches: generating functions, and symmetric polynomials using Newton's identities.

Proof I. Generating Functions

Let

$$G(x) = \sum_{k=1}^{\infty} k^3 x^k$$

denote the ordinary generating function of the cubes.

As derived in Appendix A,

$$G(x) = \frac{x(1 + 4x + x^2)}{(1-x)^4}.$$

Now let

$$S_n = \sum_{k=1}^{n} k^3$$

and consider the ordinary generating function of these partial sums:

$$F(x) = \sum_{n=1}^{\infty} S_n x^n.$$

By the definition of S_n,

$$F(x) = \sum_{n=1}^{\infty} \left(\sum_{k=1}^{n} k^3 \right) x^n.$$

Interchanging the order of summation,

$$\begin{aligned} F(x) &= \sum_{k=1}^{\infty} k^3 \sum_{n=k}^{\infty} x^n \\ &= \sum_{k=1}^{\infty} k^3 \frac{x^k}{1-x} \\ &= \frac{1}{1-x} \sum_{k=1}^{\infty} k^3 x^k. \end{aligned}$$

Recognising the inner summation as $G(x)$, we obtain

$$F(x) = \frac{G(x)}{1-x}.$$

Consequently,

$$F(x) = \frac{x(1+4x+x^2)}{(1-x)^5}.$$

Using the binomial-series expansion,

$$\frac{1}{(1-x)^5} = \sum_{r=0}^{\infty} \binom{r+4}{4} x^r,$$

we obtain

$$F(x) = x(1+4x+x^2) \sum_{r=0}^{\infty} \binom{r+4}{4} x^r.$$

Comparing coefficients of x^n,

$$S_n = \binom{n+3}{4} + 4\binom{n+2}{4} + \binom{n+1}{4},$$

where $\binom{m}{4} = 0$ for non-negative integers $m < 4$.

As shown in Appendix B, this expression simplifies to

$$S_n = \frac{n^2(n+1)^2}{4}.$$

Therefore,

$$\boxed{\sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k\right)^2}.$$

Proof II. Symmetric Polynomials and Newton's Identities

Let $x_1, x_2, \ldots, x_n$ be indeterminates. Define the elementary symmetric polynomials by

$$e_r = \sum_{1 \le i_1 < \cdots < i_r \le n} x_{i_1} x_{i_2} \cdots x_{i_r},$$

and the power sum symmetric polynomials by

$$p_r = \sum_{i=1}^{n} x_i^r.$$

The first three Newton identities are

$$p_1 = e_1,$$

$$p_2 = e_1 p_1 - 2e_2,$$

$$p_3 = e_1 p_2 - e_2 p_1 + 3e_3.$$

Consequently,

$$p_2 = e_1^2 - 2e_2$$

and

$$\boxed{p_3 = e_1^3 - 3e_1 e_2 + 3e_3}.$$

We now specialise to $x_k = k$, for $1 \leq k \leq n$.
The first elementary symmetric polynomial is

$$e_1 = \sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

The second and third elementary symmetric polynomials, as derived independently in Appendix C, are

$$e_2 = \frac{n(n+1)(n-1)(3n+2)}{24}$$

and

$$e_3 = \frac{n^2(n-1)(n-2)(n+1)^2}{48}.$$

Substituting these expressions into

$$p_3 = e_1^3 - 3e_1 e_2 + 3e_3,$$

we obtain

$$\begin{aligned} p_3 = {} & \frac{n^3(n+1)^3}{8} \\ & - \frac{n^2(n+1)^2(n-1)(3n+2)}{16} \\ & + \frac{n^2(n-1)(n-2)(n+1)^2}{16}. \end{aligned}$$

Factoring and simplifying,

$$\begin{aligned} p_3 &= \frac{n^2(n+1)^2}{16}\left[2n(n+1) - (n-1)(3n+2) + (n-1)(n-2)\right] \\ &= \frac{n^2(n+1)^2}{16} \cdot 4 \\ &= \frac{n^2(n+1)^2}{4}. \end{aligned}$$

Since $p_3 = \sum_{k=1}^{n} k^3$, it follows that

$$\boxed{\sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k\right)^2}.$$

A Derivation of the Generating Function for Cubes

Consider the geometric series

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}.$$

Define the differential operator

$$D = x\frac{d}{dx}.$$

Since

$$D(x^k) = kx^k,$$

repeated application of D gives

$$D^3\left(\sum_{k=0}^{\infty} x^k\right) = \sum_{k=1}^{\infty} k^3 x^k.$$

We evaluate the right-hand side successively:

$$D\left(\frac{1}{1-x}\right) = \frac{x}{(1-x)^2},$$

$$D^2\left(\frac{1}{1-x}\right) = \frac{x(1+x)}{(1-x)^3},$$

and

$$D^3\left(\frac{1}{1-x}\right) = \frac{x(1+4x+x^2)}{(1-x)^4}.$$

Therefore, the ordinary generating function for the cubes is

$$\boxed{G(x) = \sum_{k=1}^{\infty} k^3 x^k = \frac{x(1+4x+x^2)}{(1-x)^4}}.$$

B Simplification of the Binomial-Coefficient Expression

From Proof I, we obtained

$$S_n = \binom{n+3}{4} + 4\binom{n+2}{4} + \binom{n+1}{4}.$$

Using the factorial definition of binomial coefficients,

$$\begin{aligned} S_n = {} & \frac{(n+3)(n+2)(n+1)n}{24} \\ & + \frac{4(n+2)(n+1)n(n-1)}{24} \\ & + \frac{(n+1)n(n-1)(n-2)}{24}. \end{aligned}$$

Factoring out the common terms,

$$\begin{aligned} S_n = \frac{n(n+1)}{24}\big[& (n+3)(n+2) \\ & + 4(n+2)(n-1) \\ & + (n-1)(n-2)\big]. \end{aligned}$$

Expanding the expression inside the brackets,

$$\begin{aligned} S_n &= \frac{n(n+1)}{24}\left[n^2+5n+6+4n^2+4n-8+n^2-3n+2\right] \\ &= \frac{n(n+1)}{24}(6n^2+6n) \\ &= \frac{n^2(n+1)^2}{4}. \end{aligned}$$

Hence,

$$\boxed{S_n = \frac{n^2(n+1)^2}{4}}.$$

C Evaluation of the Second and Third Elementary Symmetric Polynomials

Let

$$P_n(t) = \prod_{k=1}^{n}(1+kt) = \sum_{r=0}^{n} e_r(n)t^r,$$

where $e_r(n)$ denotes the r-th elementary symmetric polynomial evaluated at $1, 2, \ldots, n$.

Observe that

$$P_n(t) = (1+nt)P_{n-1}(t).$$

Comparing coefficients of t^r gives the recurrence

$$e_r(n) = e_r(n-1) + ne_{r-1}(n-1).$$

Evaluation of e_2

For $r = 1$, the recurrence gives

$$e_1(n) = e_1(n-1) + n.$$

Since

$$e_1(n) = \frac{n(n+1)}{2},$$

the recurrence for e_2 becomes

$$\begin{aligned} e_2(n) &= e_2(n-1) + ne_1(n-1) \\ &= e_2(n-1) + \frac{n^2(n-1)}{2}. \end{aligned}$$

We claim that

$$e_2(n) = \frac{n(n+1)(n-1)(3n+2)}{24}.$$

To verify this expression, let

$$E_2(n) = \frac{n(n+1)(n-1)(3n+2)}{24}.$$

For $n \geq 2$,

$$\begin{aligned} E_2(n) - E_2(n-1) &= \frac{n(n-1)}{24}\left[(n+1)(3n+2) - (n-2)(3n-1)\right] \\ &= \frac{n^2(n-1)}{2} \\ &= ne_1(n-1). \end{aligned}$$

Furthermore, $E_2(1) = 0 = e_2(1)$. Hence, by the recurrence,

$$\boxed{e_2(n) = \frac{n(n+1)(n-1)(3n+2)}{24}}.$$

Evaluation of e_3

For $r = 3$, the recurrence gives

$$e_3(n) = e_3(n-1) + ne_2(n-1).$$

Using the expression for $e_2(n-1)$,

$$e_2(n-1) = \frac{n(n-1)(n-2)(3n-1)}{24},$$

we obtain

$$e_3(n) = e_3(n-1) + \frac{n^2(n-1)(n-2)(3n-1)}{24}.$$

Since $e_3(2) = 0$, we claim that

$$e_3(n) = \frac{n^2(n-1)(n-2)(n+1)^2}{48}.$$

Let

$$E_3(n) = \frac{n^2(n-1)(n-2)(n+1)^2}{48}.$$

We verify this expression using the recurrence. For $n \geq 3$,

$$\begin{aligned} E_3(n) - E_3(n-1) &= \frac{n^2(n-2)}{48}\left[(n-1)(n+1)^2 - (n-1)^2(n-3)\right] \\ &= \frac{n^2(n-1)(n-2)}{48}\left[(n+1)^2 - (n-1)(n-3)\right] \\ &= \frac{n^2(n-1)(n-2)(3n-1)}{24} \\ &= ne_2(n-1). \end{aligned}$$

Furthermore, $E_3(2) = 0 = e_3(2)$. Hence, by the recurrence,

$$\boxed{e_3(n) = \frac{n^2(n-1)(n-2)(n+1)^2}{48}}.$$